Is it possible to do a computational postcolonial literature project?

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The field of digital humanities, and within that, computational literary studies, has laudably grown to include literary traditions across multiple languages and countries. The field's Anglocentrism has been questioned (i.e. Bailey 2012 and Fiormonte 2016) and manifestos written to widen access for people of color and the Global South (recent work include Baeza et al. 2023, Frazier et al. 2023, Chapman 2023, Quintanilla and Horcasitas 2023). Even so, digital humanists continue to ask important questions about access to digital materials, digital infrastructures, and to the field of digital humanities itself.

This presentation asks those same questions when it comes to the inclusion, or lack thereof, of literary traditions that are commonly associated with the rubric of Anglophone postcolonial literature, typically encompassing South Asian, Caribbean, and African literature. Due to the after-effects of colonial rule, including its more sustained influence on culture, these literatures often lack access to the crucial resources and infrastructures that determine their representation within the publishing industry and academia. How have these inequities been replicated and perpetuated in the digital realm? What steps can we take to ensure that the field of digital humanities, particularly computational literary studies, does not follow the same trajectory?

Roopika Risam writes in New Digital Worlds: Postcolonial Digital Humanities in Theory, Praxis, and Pedagogy that "postcolonial digital humanities is not only a theoretical or analytical approach to the digital cultural record. Rather, it requires praxis at the intersection of digital technologies and humanistic inquiry: designing new workflows and building new archives, tools, databases, and other digital

objects that actively resist reinscriptions of colonialism and neocolonialism" (4). In the same spirit, this presentation identifies how reinscriptions of colonialism and neocolonialism in the digital realm have engendered obstacles to the inclusion of postcolonial literature in computational literary studies and digital humanities (also touched on in Priego 2022 and Risam 2024). It does so based on the experiences and failures of three different attempts at a computational postcolonial literature project. I discuss how far I have managed to push these projects despite the bibliographic, digital archiving, metadata, and infrastructural issues faced, and consider different ways to circumvent them.

The first attempt highlights the unique difficulty of finding or creating bibliographies of postcolonial literature as a basis for corpus construction. Most literary traditions under the postcolonial rubric have not benefited from the work of as many librarians and archivists as more established traditions like British or American literature (see Smith and Whearty 2023 on the invisible labor of digitization). Comprehensive lists of Zimbabwean literature, for example, are hard to find. Attempting to create such a bibliography only solves half the problem, as there is very little consensus as to what such a bibliography should look like. Tackling this problem by including as many texts as possible instead runs into the issue of a lack of documentation of published texts. Despite these complexities, I chose the comprehensive listing route and ended up with 1,962 fiction texts written in or translated into English from five different regions: Caribbean, African, South and Southeast Asian, Pacific Islands, and Latin America. However, we then ran into the second barrier: only 112 of those texts were obtainable online, which is about 5% of the texts listed. This highlights another major obstacle to the inclusion of postcolonial literature into the field of digital humanities — the lack of digitally available and archived texts. The likely

scarce resources dedicated to digitizing postcolonial literature are further encumbered by sparse depositories and archives of the physical books, many of which probably had smaller print runs and distribution networks.

For my second attempt, I narrowed my research question considerably and located a digital corpora of texts based on a single publisher: the Heinemann African Writers Series (AWS). This series is well known — with authors like Chinua Achebe, Wole Soyinka, and Ngũgĩ wa Thiong'o among its ranks — and long-running (1962-2003), which means that the print runs are bigger, the bibliography reasonably populated (359 texts), and the books better digitized. However, this approach faced a third barrier; that is — monopolization — reducing research access to the already limited digital corpora of postcolonial literature. Furthermore, ownership of digitized postcolonial corpora is monopolized by multinational companies located in previous imperial centers, thus reinscribing colonial dynamics. The AWS corpus was originally compiled by Chadwyck Healey, which was acquired by ProQuest in 1999. As a researcher working in a well-funded university, its well-funded library was still unable to justify the cost of a yearly subscription to the large bundle that contains the AWS corpus. ProQuest does not offer the option of purchasing access only to the series, and if purchased, the corpus would only be available for analysis through ProQuest's proprietary platform.

This monopolization problem continued with the third attempt. This time, I turned to any related corpora that the library already owns. Luckily, I was able to request access to the Caribbean Literature corpora as originally compiled by Alexander Street, which of course, was acquired by ProQuest in 2016. Even though the library owned the corpora (as opposed to merely leasing or subscribing), ProQuest still needed 4 months to provide access to the raw text files. After obtaining the files, another

obstacle to the possibility of computational postcolonial literature projects became apparent: lack of attention and resources to metadata, a symptom of the same problem of lack of resources mentioned earlier. The metadata file provided was entirely empty except for a column, detailing inaccurately, the language of the text. The file organization also made it difficult to understand what was actually in the corpus, which the presentation will elaborate on. After a considerable amount of time spent decoding the files, the corpus only contained less than 50% English texts, which is 1 among 18 languages within the corpus. This highlights yet another complication — that of inherent multilingualism, and especially for languages underserved in the digital sphere like Swahili — for digital humanists that are interested in postcolonial literature. A consideration of the literatures of South Asia or Africa, for example, beyond the Anglophone, would be inhibited by the lack of support in Unicode (Hossain 2024), much less NLP, in their many indigenous languages. However, developments such as NLP applications for Igala (Ayegba et al. 2017) and more recently, natural language generation for 517 African languages (Adebara et al. 2024) have moved the field forward.

I end the presentation by discussing possible ways forward (such as by using found or limited corpora) despite the many barriers that my attempts have encountered, and by calling for more resources within the field of digital humanities to be dedicated to building digital access and infrastructure for postcolonial literature, as well as including more postcolonial studies scholars within the field. While the challenges described are not unique to the computational analyses of postcolonial literature, they disproportionately impact the already under-supported area of postcolonial literature, reinscribing the colonial and neo-

colonial dynamics that have characterized it.

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